Fabio Matheus Mantelli

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Professional Summary

DevOps/Infrastructure Engineer and Electrical Engineer with 15+ years operating and maintaining large-scale monitoring infrastructure across international environments. Hands-on experience with Linux/Windows Server administration, VPN configuration, Python scripting, CI/CD concepts, and cloud platforms (AWS). Deployed Zabbix across an 8-country network monitoring 100+ real-time phasor measurement devices. AWS Certified AI Practitioner. U.S. permanent resident authorized to work without sponsorship.

Core Competencies

Infrastructure	Linux, Windows Server, VPN configuration, TCP/IP networking
Monitoring & Observability	Zabbix, Grafana, openHistorian, real-time telemetry pipelines
Scripting & Automation	Python, Bash, C/C++
Databases	PostgreSQL, MySQL, InfluxDB, SNAPdb
Cloud	AWS (EC2, S3), AWS Certified AI Practitioner
Software Design	OOP principles, software architecture (applied studies)

Professional Experience

Infrastructure & Data Engineer *Nov 2023 – Present*
INESC P&D Brasil (Remote)

- Build and maintain Python-based data integration pipelines for real-time telemetry ingestion and validation across distributed measurement infrastructure.
- Administer Linux environments supporting continuous data acquisition from high-voltage transmission monitoring systems.
- Validate telemetry data quality and pipeline integrity, ensuring reliable delivery of real-time operational data for engineering analysis.
- Design and implement post-event analysis tooling to support infrastructure performance reporting and incident review.
- Contributed to applied research involving synchrophasor-based protection analytics and real-time telemetry processing for interconnected AC/DC power systems with embedded HVDC infrastructure.

Infrastructure Engineer — International Monitoring Platform *Jun 2015 – Oct 2023*
FEESC / MedFasee Program — Brazil

- Deployed and operated Zabbix monitoring across an 8-country international infrastructure (Brazil, Argentina, Chile, Uruguay, Portugal, Spain, Italy, Croatia), supervising 100+ synchronized phasor measurement devices.
- Administered Windows Server and Linux environments supporting 24/7 real-time data collection, concentrator services, and time-synchronized telemetry pipelines.
- Configured VPN tunnels and secure communication channels for inter-site data exchange between measurement stations and central data centers.
- Managed PostgreSQL and InfluxDB databases for time-series telemetry storage; performed data completeness tracking, latency validation, and capacity monitoring.
- Coordinated device commissioning and infrastructure expansion across remote substations.

Research Engineer — Monitoring Infrastructure & Software Development *Sep 2009 – May 2015*
Federal University of Santa Catarina (UFSC) — Brazil

- Developed C/C++ analytical tooling applying OOP design principles for real-time data processing and disturbance detection in synchrophasor infrastructure (undergraduate thesis).
- Deployed and maintained PMU-based monitoring infrastructure at transmission substations, including hardware commissioning, software configuration, and data pipeline validation.
- Built MATLAB-based analysis tools for time-series signal processing and infrastructure performance evaluation.

Selected Infrastructure Contributions

- **Zabbix International Deployment:** Designed and operated Zabbix monitoring across an 8-country network (Europe and South America) supervising 100+ real-time phasor measurement devices with alerting, availability tracking, and performance dashboards.
- **openPDC/openHistorian Deployment — CTEEP (2022):** Delivered full technical implementation of the Grid Protection Alliance open-source monitoring stack (openPDC, openHistorian 2, openXDA) at CTEEP, the largest transmission

utility in São Paulo state.

- **Multi-Site VPN & Network Infrastructure:** Configured and maintained secure VPN tunnels and TCP/IP networking for real-time inter-site data collection across geographically distributed measurement stations.
- **Python Telemetry Pipeline:** Designed data integration pipelines in Python for telemetry ingestion, quality validation, and operational reporting from distributed infrastructure endpoints.
- **Real-Time Protection Analytics Infrastructure:** Supported telemetry and data-processing infrastructure used in synchrophasor-based protection and transient instability studies for interconnected AC/DC transmission systems.

Technical Publications

- **Electric Power Systems Research (Qualis A1), 2026:** Co-author of publication involving synchrophasor-based backup protection analytics and transient instability mitigation in HVDC-integrated transmission systems.
- **Power Systems Computation Conference (PSCC), 2026:** International conference publication focused on real-time synchrophasor analytics and protection strategies for interconnected AC/DC systems.

Education

Bachelor of Electrical Engineering *Aug 2020*
Federal Institute of Santa Catarina — Brazil

Graduate-Level Coursework in Electrical Engineering — Power Systems *2012 – 2013*
Federal University of Santa Catarina (UFSC) — Brazil
27 graduate credits: Power System Dynamics, Operation Planning, Control Systems, Numerical Optimization

Associate Degree in Energy Systems *Oct 2011*
Federal Institute of Santa Catarina — Brazil

Certifications

- **AWS Certified AI Practitioner** — Amazon Web Services, 2025
- **AWS Certified Solutions Architect – Associate** (Exam Scheduled for July 2026)

Technical Skills

Systems	Linux, Windows Server, VPN configuration, TCP/IP networking
Monitoring	Zabbix, Grafana, openHistorian, openPDC
Cloud	AWS (EC2, S3)
Databases	PostgreSQL, MySQL, InfluxDB, SNAPdb
Programming	Python, C/C++, MATLAB
Domain Knowledge	Electrical Engineering, Power Systems, IEEE C37.118
Languages	English (Professional), Portuguese (Native)